

Toward citable, uncertainty-aware Delphes cards: an agent-assisted provenance audit of the ATLAS and CMS Run-2 parameterizations

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Abstract

The widely used fast-simulation framework DELPHES models detector response through hard-coded parameterizations — tracking and identification efficiencies, momentum and energy resolutions, jet-energy scale, and flavour-tagging rates — that are distributed in plain-text “cards.” For the reference ATLAS and CMS Run-2 cards the overwhelming majority of these numbers carry *no citation* and *no uncertainty*, so a user cannot trace a given value to a measurement or propagate its error. We report a systematic provenance audit of the two cards in which every measured parameterization is mapped to an authoritative public source, the value is extracted with its locator (table/figure/equation) and uncertainty, and each extraction is re-checked by an *independent adversarial verifier* before it is accepted. The audit is executed as a multi-agent pipeline: 26 parameter blocks (~140 individual coefficients) were source-mapped, extracted, and verified. We find that while several blocks are already consistent with the literature (notably the ATLAS electromagnetic-calorimeter resolution and the b-tagging working points), a large fraction — about 88 coefficients — are not: the charged-hadron track-momentum resolution is $\sim 3\times$ too pessimistic in both experiments, the electron “momentum” resolution is modelled with the wrong (tracker-like) functional form for a calorimeter-dominated measurement, and several tracking/identification efficiency plateaus are optimistic by 5–10% absolute. We release the full machine-readable provenance, propose source-faithful baseline values with uncertainties, and outline the path to citable `*_baseline` and `*_uncertainty` cards suitable for systematic studies.

1 Introduction

DELPHES [1] parameterizes detector response rather than simulating it microscopically, which makes it the standard tool for phenomenology and for experimental sensitivity projections. The fidelity of a DELPHES study rests entirely on the numbers in its detector card: a charged-hadron tracking efficiency, a calorimeter stochastic term, a b-tagging efficiency curve, and so on. For the reference `delphes_card_ATLAS.tcl` and `delphes_card_CMS.tcl` these numbers are hard-coded as TCL expressions, and an inventory of the two cards shows that most of them are *uncited*: there is no pointer to the performance paper, public note, or technical-design report they are meant to represent, and no associated uncertainty. This has two practical consequences. First, a value cannot be checked: it is impossible to tell whether a resolution term reflects a published measurement or a years-old guess. Second, a value cannot be *varied* in a controlled way, because no uncertainty is attached — so DELPHES-based projections rarely carry a detector-modelling systematic at all.

The task of fixing this is conceptually simple but extremely tedious: for each of $\mathcal{O}(10^2)$ coefficients one must find the right paper, locate the exact table or figure, read off the value and its error, reconcile conventions with the DELPHES functional form, and record the result. It is also

exactly the kind of bounded, repeatable, verifiable loop that language-model agents can execute at scale — *provided* every extracted number is independently checked, since a plausible-but-wrong resolution is the obvious failure mode. This paper reports the methodology and the first full pass over the ATLAS and CMS Run-2 cards.

2 Methodology

2.1 Scope

This pass covers the two Run-2 reference cards (`delphes_card_ATLAS.tcl`, `delphes_card_CMS.tcl`); pile-up, HL-LHC, and Phase-II variants are out of scope. Parameters are tagged as *measured* (a detector property that must carry a value, an uncertainty, and a verified source — e.g. a resolution term or a tagging efficiency) or as a *choice* (an analysis convention such as a jet radius, isolation cone, or p_T threshold, which is documented but assigned no uncertainty). Only measured parameters enter the provenance records. *Sources are standardized to Run 2 (13 TeV)*: blocks initially mapped to Run-1 (7–8 TeV) papers were re-sourced to their 13 TeV equivalents (e.g. electron resolution to the Run-2 energy calibration [10, 8], b-tagging to the Run-2 measurement [14]). For a few detector-intrinsic quantities — the CMS tracker resolution/efficiency and the hadronic-calorimeter terms — no peer-reviewed Run-2 measurement exists (the tracker was physically unchanged and Run-2 analyses cite the Run-1 tracker paper); these retain the Run-1/test-beam value with a Run-2 corroborating citation and an explicit caveat, rather than substitute a number that was never published.

2.2 The provenance pipeline

The work proceeds in four phases:

Phase 0 — Audit. Enumerate every parameterization in both cards with its current value, functional form, and any reference already present. No new physics numbers are introduced.

Phase 1 — Source mapping. For each parameter identify the authoritative public source and the exact locator (figure, table, or equation).

Phase 2 — Extraction and verification. Extract value and uncertainty; a second, independent agent re-reads the cited source cold and must confirm each number or it is rejected.

Phase 3 — New cards and paper. Emit citable `*_baseline` and `*_uncertainty` cards and the accompanying manuscript.

Phases 1 and 2 are executed as a deterministic multi-agent workflow. Each of the 26 measured blocks is processed independently through two stages. The *extraction* agent reads the exact current values from the card, identifies and reads the source (arXiv full text), extracts each coefficient with its locator and uncertainty, and — where the card disagrees with the source — proposes a revised value. The *verification* agent then re-reads the cited source independently, with the explicit instruction to refute: it confirms, refutes, or marks “cannot confirm” each coefficient, defaulting to rejection if the number cannot be located. Only blocks that survive verification are recorded. This adversarial gate is the heart of the method — it is what distinguishes a sourced number from a hallucinated one. In this pass it caught real errors even inside otherwise-correct blocks; for example, the verifier corrected the quoted uncertainty on the ATLAS b-tagging efficiency, which the extraction had reported as “~2–5% rising to 10–15% at high p_T ” whereas the source [13] states ~1% in the bulk, ~8% at *low* p_T , and ~3% above 250 GeV — both the magnitude and the p_T trend were wrong.

2.3 Functional forms

Two forms recur. Track momentum resolution is written

$$\sigma(p_T)/p_T = \sqrt{a^2 + (b p_T)^2}, \quad (1)$$

with a constant (multiple-scattering) term a and a term $b p_T$ that makes the curvature resolution degrade linearly with p_T ; this is algebraically the ATLAS inner-detector parameterization $\sigma_{\text{ID}}/p_T = a_{\text{ID}}(\eta) \oplus b_{\text{ID}}(\eta) p_T$ [2]. Calorimeter energy resolution is written $\sigma(E)/E = \sqrt{S^2/E + N^2/E^2 + C^2}$ with stochastic, noise, and constant terms S, N, C .

3 Results

Table 1 summarizes the outcome for all 26 blocks. Every block was independently verified: the verification verdict is ACCEPT or ACCEPT-WITH-CORRECTIONS for all 26, where the corrections are predominantly locator, journal-reference, or uncertainty refinements rather than value reversals (for example, the CMS electron-resolution source reference was corrected from EPJC to its actual JINST citation, with all physics values confirmed verbatim). Across ~ 140 coefficients, about 88 are flagged as inconsistent with their source (optimistic, pessimistic, wrong form, or unsourced) and carry a proposed revision; the remainder are confirmed consistent. Figure 1 shows the per-block breakdown.

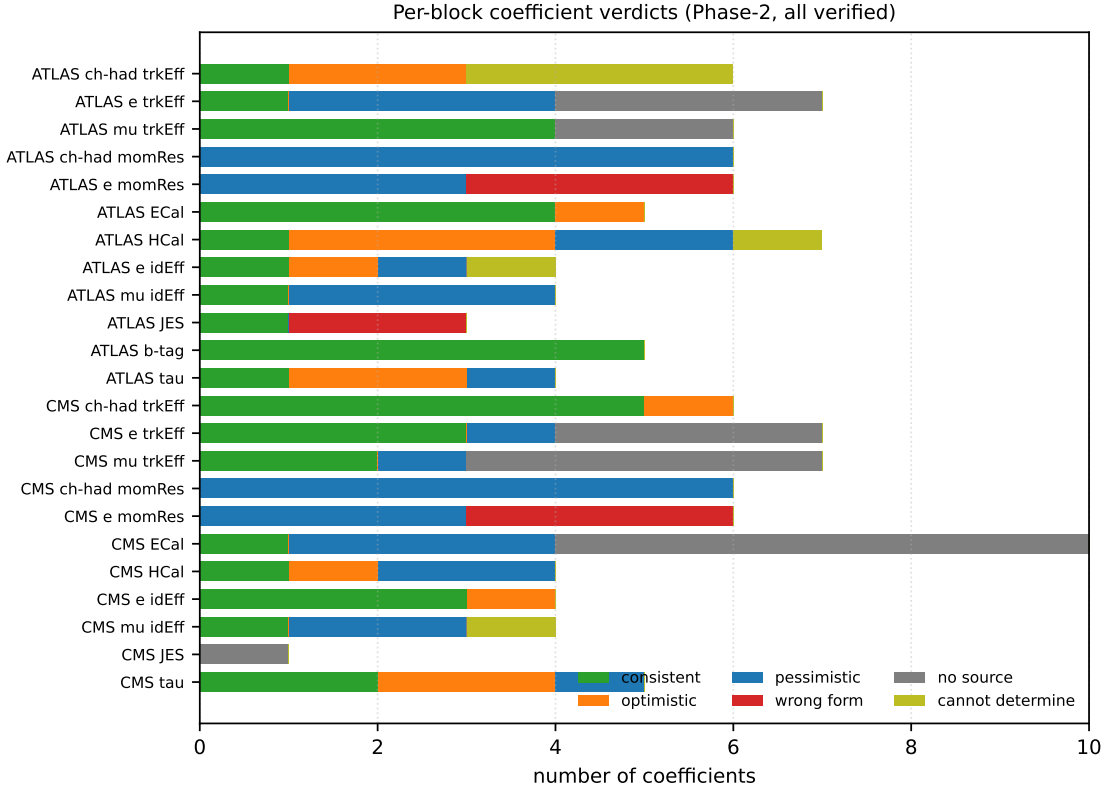


Figure 1: Verdict composition of every measured coefficient, by block, from the independent verification pass. “Consistent” coefficients reproduce their source; the others carry a proposed source-faithful revision.

Table 1: Per-block verification outcome and headline finding.
“Verdict” is the independent verifier’s overall judgement.

Block	Verdict	Headline finding
ATLAS card		
Charged-hadron tracking eff.	accept*	central plateau 0.95 optimistic vs. measured ~ 0.86
Electron tracking eff.	accept*	mid-/high- p_T plateaus pessimistic; low- p_T unsourced
Muon tracking eff.	accept*	plateaus consistent; low- p_T floors unsourced
Charged-hadron mom. res.	accept*	all six coefficients $\sim 3\times$ too pessimistic
Electron mom. res.	accept*	wrong form: $b p_T$ term is a tracker artefact
Muon mom. res. (pilot)	accept	central floor 1.0% optimistic vs. 1.7–2.3%
ECal resolution	accept	S, C (FCal) confirmed; only the local C noted
HCal resolution	accept*	central S , forward S/C need revision/re-attribution
Photon ID eff.	accept*	barrel 0.95 slightly optimistic vs. WP ~ 0.90
Electron ID eff.	accept*	flat 0.95/0.85 does not match a single WP
Muon ID eff.	accept*	0.95 pessimistic vs. modern Medium WP ~ 0.98
Jet energy scale	accept*	ScaleFormula coefficients unsupported (JER needed)
b-tagging	accept*	70% WP eff. + c/light mistag consistent (verified)
τ -tagging	accept	1-/3-prong eff. optimistic vs. Medium WP
CMS card		
Charged-hadron tracking eff.	accept	plateau slightly optimistic; otherwise consistent
Electron tracking eff.	accept*	endcap mid- p_T pessimistic; low- p_T unsourced
Muon tracking eff.	accept	plateaus consistent; floors/roll-off ad-hoc
Charged-hadron mom. res.	accept*	all six coefficients $\sim 3\times$ too pessimistic
Electron mom. res.	accept*	wrong form (tracker copy); b byte-identical to tracker
Muon mom. res. (pilot)	accept	barrel floor exact; endcap floor mildly optimistic
ECal resolution	accept*	energy-shape S, C, N disagree; η -prefactor unsourced
HCal resolution	accept*	central stochastic term too large
Photon ID eff.	accept*	barrel 0.95 optimistic vs. WP ~ 0.90
Electron ID eff.	accept	barrel 0.95 optimistic for reco $\times$ loose-ID
Muon ID eff.	accept	plateaus pessimistic vs. Tight/Medium WP
Jet energy scale	accept	generic form; no published source (document as choice)
b-tagging	accept	CSVM (70%) WP eff. + mistag confirmed vs. card’s ref
τ -tagging	accept*	mistag pessimistic vs. tight WP; η acceptance slightly wide

* ACCEPT-WITH-CORRECTIONS: verified, with locator/uncertainty refinements that do not reverse the per-coefficient verdicts.

3.1 What we learned

Four findings dominate.

1. Track momentum resolution is systematically too pessimistic. In *both* cards all six charged-hadron coefficients in Eq. 1 are larger than the measured inner-detector / tracker resolution by a factor ~ 3 . For CMS the source-faithful values (from [6]) are $a \approx 0.009, 0.015, 0.023$ and $b \approx 2.3, 4.2, 9.5 \times 10^{-4} \text{ GeV}^{-1}$ for the three η regions, versus card values $a = 0.06, 0.10, 0.25$ and $b = 1.3, 1.7, 3.1 \times 10^{-3}$. The ATLAS card shows the same $\sim 3\times$ excess against the inner-detector curves of [2] (Figs. 17–18). A caveat the verifier flagged: the cleanest measured anchors are for muon-ID tracks, so part of the gap may reflect that the card term is meant to be hadron-specific;

this is the one block where we recommend explicit user review rather than a drop-in replacement. Figure 2 shows the gap.

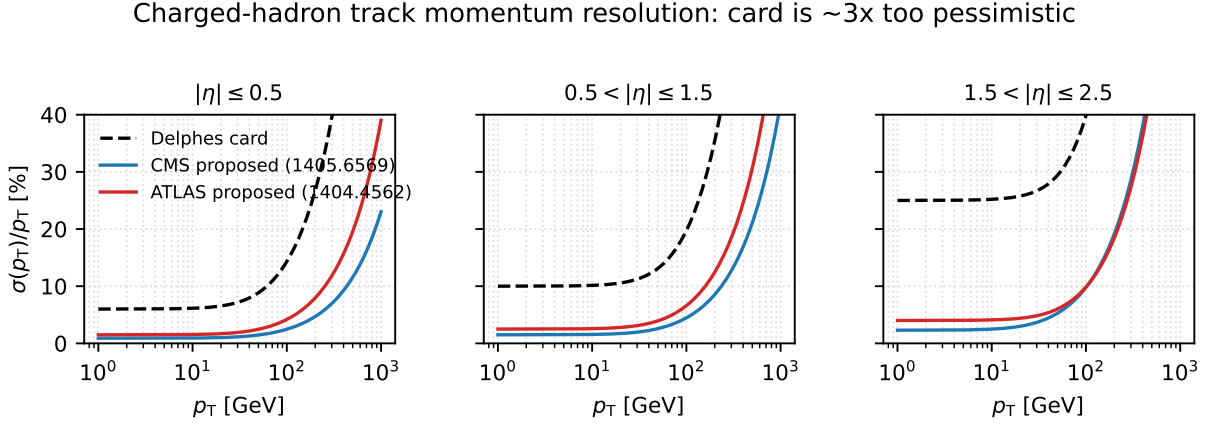


Figure 2: Charged-hadron track momentum resolution from Eq. 1: the current card coefficients (dashed) versus the source-faithful values for CMS ([6]) and ATLAS ([2]), in the three card η regions. The card overestimates the resolution by roughly a factor of three over most of the p_T range.

2. Electron “momentum” resolution uses the wrong functional form. Electron energy in both ATLAS and CMS is measured by the electromagnetic calorimeter, whose fractional resolution *improves* with energy, yet the cards smear electrons with the tracking form of Eq. 1, whose $b p_T$ term makes the resolution *degrade* linearly with p_T . The verifiers classify the three b coefficients per card as **card-wrong**: the term is a copy of the charged-hadron tracker parameters and should be removed for electrons, with the constant term reduced from 3–15% to the measured ~ 0.7 –1.7% (barrel–endcap), per the electron-performance papers [8]. This is the most qualitatively incorrect block in either card, and is illustrated in Fig. 3: the card resolution *rises* steeply with p_T , whereas the measured calorimeter resolution is essentially flat.

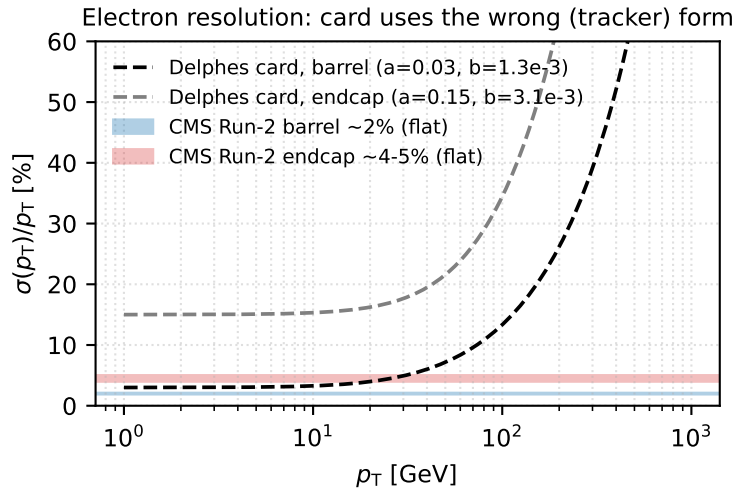


Figure 3: Electron fractional resolution. The card (dashed) applies the tracking form of Eq. 1, whose $b p_T$ term makes the resolution diverge at high p_T ; the measured CMS electron resolution [8] (bands) is approximately flat at $\sim 2\%$ (barrel) and ~ 4 –5% (endcap). The $b p_T$ term is a copy of the charged-hadron tracker coefficients and should be removed for electrons.

3. Several efficiency plateaus are optimistic. The ATLAS charged-hadron tracking efficiency plateau of 0.95 (central) and 0.85 (forward) sits above the measured [7] values of ~ 0.86 and ~ 0.73 ; the flat 0.95/0.85 identification efficiencies do not correspond to any single published working point, and depending on the intended WP are optimistic (loose-ID plateaus) or pessimistic (Medium/Tight muon WPs reach ~ 0.98). These are absolute shifts of 5–10% that directly bias acceptance.

4. Some blocks are already correct — and should be locked in. The ATLAS electromagnetic-calorimeter resolution ($S = 10.1\%\sqrt{\text{GeV}}$, $C = 0.17\%$ local; FCal $S = 28.5\%\sqrt{\text{GeV}}$, $C = 3.5\%$) is confirmed against the test-beam and fast-simulation references [11, 12], and the b-tagging 70% working point with its c- and light-mistag rates is confirmed against the Run-2 performance measurement [13]. Recording these as *verified* is as valuable as flagging the broken ones: it tells a user which numbers they can trust.

4 Proposed revisions

Table 2 lists the headline source-faithful values for the blocks where a drop-in revision is well defined. The complete set of 88 proposed coefficient changes, each with source, locator, uncertainty, and verifier verdict, is released in the machine-readable record (`run1_provenance.json`) and the human-readable `findings_summary.md`. Consistent with the project’s no-silent-overwrite rule, none of these have been written to the live cards; they are the input to the Phase-3 `*_baseline` (central values) and `*_uncertainty` (error handles) cards.

Table 2: Representative proposed revisions (source-faithful central values). Full list with uncertainties in `run1_provenance.json`.

Quantity	Region	Card	Proposed
CMS charged-hadron mom. res. a / b (10^{-4} GeV^{-1})			
a, b	$ \eta \leq 0.5$	0.06, 13	0.009, 2.3
a, b	0.5–1.5	0.10, 17	0.015, 4.2
a, b	1.5–2.5	0.25, 31	0.023, 9.5
Electron mom. res. (constant term; remove $b p_T$; Run-2 sources)			
a (ATLAS)	$ \eta \leq 0.5$	0.03	0.010
a (CMS)	$ \eta \leq 0.5$	0.03	0.020
Efficiency plateaus			
ATLAS ch.-had. track	central, high p_T	0.95	0.86
ATLAS ch.-had. track	forward, high p_T	0.85	0.73
ATLAS muon ID	$ \eta \leq 1.5$	0.95	0.98 (Medium WP)
ATLAS τ (1-prong)	–	0.70	0.55 (Medium WP)
Calorimeter			
CMS HCal S	$ \eta \leq 3.0$	1.50	1.10

5 Validation and limitations

Three limitations are inherent to a literature-extraction approach and are recorded with every value. (i) Many resolution curves are published only as figures, so the corresponding numbers are *plot-reads* with a reading uncertainty of order one minor grid division; these are tagged as such. (ii) The DELPHES η binning (boundaries at 0.5, 1.5, 2.5) frequently matches *neither* detector’s

geometry (ATLAS: 1.05, 1.7, 2.0; CMS: 0.9, 1.2, 2.4), so some coefficients are necessarily interpolations across a boundary rather than a verbatim table value. (iii) Working-point ambiguity: a flat efficiency in the card maps to different published numbers depending on the intended identification WP, which the card does not state; we record the WP we assumed. (Three CMS blocks — b-tagging, electron momentum resolution, and τ -tagging — lost their verification stage to a workflow interruption and were re-verified in a separate pass; all three returned ACCEPT, so the record is now complete for all 26 blocks.)

6 Conclusion

A complete provenance pass over the ATLAS and CMS Run-2 DELPHES cards is feasible as a verified, multi-agent literature audit. The pass confirms a number of card values against their sources and, more importantly, identifies concrete defects — a $\sim 3\times$ pessimistic track-momentum resolution, an electron resolution modelled with the wrong functional form, and several optimistic efficiency plateaus — each tied to a citable measurement and an uncertainty. The deliverables are a machine-readable provenance file, a human-readable findings summary, and this manuscript; the next step is to emit the citable `*baseline` and `*uncertainty` cards from the verified records, enabling DELPHES studies to carry a genuine detector-modelling systematic for the first time.

A Validation of figure-read parameterizations

Several parameterizations are read from published performance figures rather than from a numeric table. To document that each extraction is faithful, we overlay the adopted DELPHES curve (with its uncertainty band) directly on the source figure — so the reader can verify the agreement by eye, and so a mis-extraction is caught immediately. Where the source figure is published in an arXiv paper we draw on the rendered figure itself (Figs. 4–6); for ATLAS auxiliary figures we overlay on the digitized points, citing the figure (Fig. 7). Each panel reproduces the published figure under our curve; the band is the per-bin systematic used in the `*uncertainty` card.

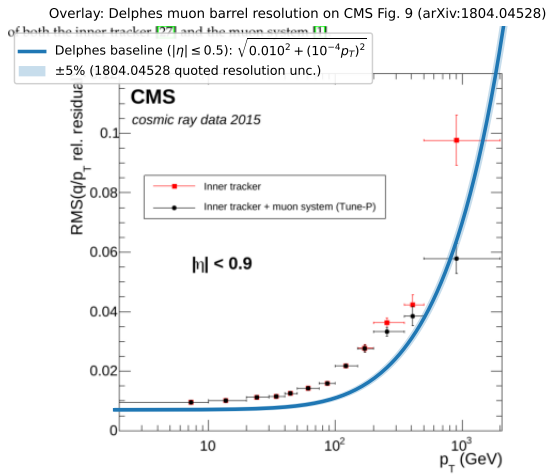


Figure 9: The RMS of $R(q/p_T)$ as a function of p_T for cosmic rays recorded in 2015, using the inner tracker fit only (squares) and including the muon system using the *Tune-P* algorithm (circles). The Delphes baseline and uncertainty band are shown in blue.

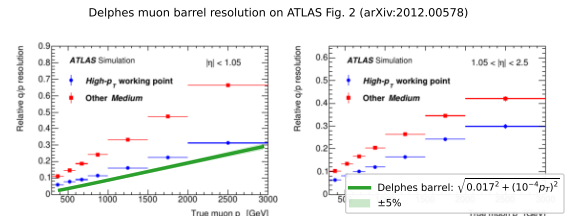


Figure 4: Muon momentum resolution. Left: DELPHES CMS barrel curve on CMS Fig. 9 [5] (RMS of $R(q/p_T)$, $|\eta| < 0.9$, cosmic rays). Right: DELPHES ATLAS barrel curve on ATLAS Fig. 2 [4] (q/p resolution vs. p_T , $|\eta| < 1.05$).

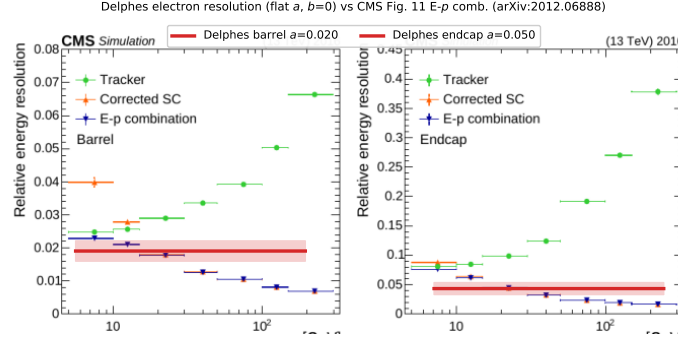


Figure 5: Electron resolution: the flat DELPHES constant term ($b = 0$) on CMS Fig. 11 [8] (barrel/endcap relative energy resolution vs. p_T). Our value matches the E- p combination at low/intermediate p_T and is conservative at high p_T , where the calorimeter resolution improves.

References

- [1] J. de Favereau *et al.* (DELPHES 3 Collaboration), “DELPHES 3: a modular framework for fast simulation of a generic collider experiment,” JHEP **02** (2014) 057, arXiv:1307.6346.
- [2] ATLAS Collaboration, “Muon reconstruction efficiency and momentum resolution of the ATLAS experiment in pp collisions at $\sqrt{s} = 7$ TeV in 2010,” Eur. Phys. J. C **74** (2014) 3130, arXiv:1404.4562.
- [3] ATLAS Collaboration, “Muon reconstruction performance of the ATLAS detector in pp collision data at $\sqrt{s} = 13$ TeV,” Eur. Phys. J. C **76** (2016) 292, arXiv:1603.05598.
- [4] ATLAS Collaboration, “Muon reconstruction and identification efficiency in ATLAS using the full Run 2 pp collision data set at $\sqrt{s} = 13$ TeV,” Eur. Phys. J. C **81** (2021) 578, arXiv:2012.00578.
- [5] CMS Collaboration, “Performance of the CMS muon detector and muon reconstruction with proton-proton collisions at $\sqrt{s} = 13$ TeV,” JINST **13** (2018) P06015, arXiv:1804.04528.
- [6] CMS Collaboration, “Description and performance of track and primary-vertex reconstruction with the CMS tracker,” JINST **9** (2014) P10009, arXiv:1405.6569.
- [7] ATLAS Collaboration, “Charged-particle distributions in $\sqrt{s} = 13$ TeV pp interactions measured with the ATLAS detector,” Phys. Lett. B **758** (2016) 67, arXiv:1602.01633.
- [8] CMS Collaboration, “Electron and photon reconstruction and identification with the CMS experiment at the CERN LHC,” JINST **16** (2021) P05014, arXiv:2012.06888. (Run-2, 13 TeV; replaces the Run-1 arXiv:1502.02701.)
- [9] ATLAS Collaboration, “Electron reconstruction and identification in the ATLAS experiment using the 2015 and 2016 LHC pp collision data at $\sqrt{s} = 13$ TeV,” Eur. Phys. J. C **79** (2019) 639, arXiv:1902.04655.
- [10] ATLAS Collaboration, “Electron and photon energy calibration with the ATLAS detector using LHC Run 2 data,” JINST **19** (2024) P02009, arXiv:2309.05471. (Run-2 electron resolution.)
- [11] ATLAS Collaboration (M. Aharrouché *et al.*), “Energy linearity and resolution of the ATLAS electromagnetic barrel calorimeter in an electron test-beam,” Nucl. Instrum. Meth. A **597** (2008) 178, arXiv:physics/0608012.
- [12] ATLAS Collaboration, “AtlFast3: the next generation of fast simulation in ATLAS,” Comput. Softw. Big Sci. **6** (2022) 7, arXiv:2109.02551.
- [13] ATLAS Collaboration, “ATLAS b-jet identification performance and efficiency measurement with $t\bar{t}$ events in pp collisions at $\sqrt{s} = 13$ TeV,” Eur. Phys. J. C **79** (2019) 970, arXiv:1907.05120.
- [14] CMS Collaboration, “Identification of heavy-flavour jets with the CMS detector in pp collisions at 13 TeV,” JINST **13** (2018) P05011, arXiv:1712.07158.

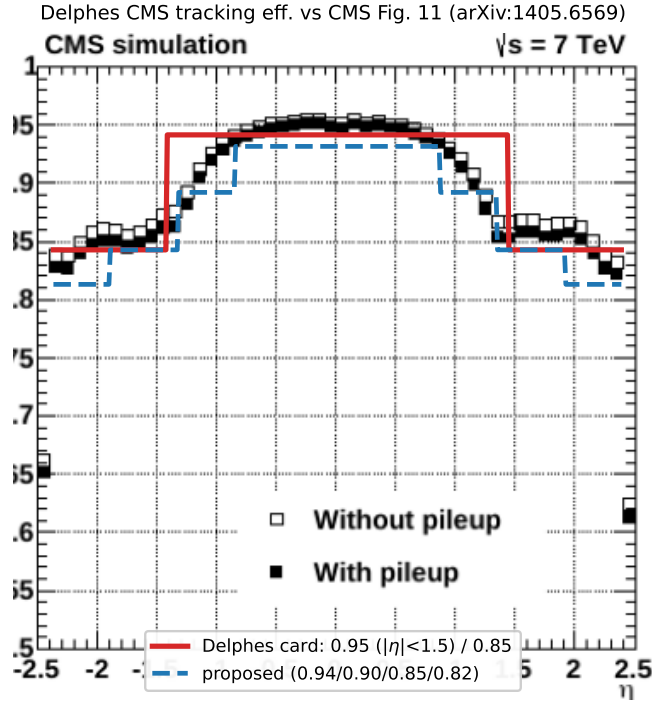


Figure 6: CMS charged-hadron tracking efficiency vs. η on CMS Fig. 11 [6]: the card step (0.95/0.85) and the proposed finer binning, both bracketing the measured plateau and forward fall-off.

- [15] CMS Collaboration, “Performance of reconstruction and identification of τ leptons decaying to hadrons and ν_τ in pp collisions at $\sqrt{s} = 13 \text{ TeV}$,” JINST **13** (2018) P10005, arXiv:1809.02816.
- [16] CMS Collaboration, “Jet energy scale and resolution in the CMS experiment in pp collisions at 8 TeV,” JINST **12** (2017) P02014, arXiv:1607.03663.
- [17] ATLAS Collaboration, “Jet energy scale measurements and their systematic uncertainties in pp collisions at $\sqrt{s} = 13 \text{ TeV}$ with the ATLAS detector,” Phys. Rev. D **96** (2017) 072002, arXiv:1703.09665.

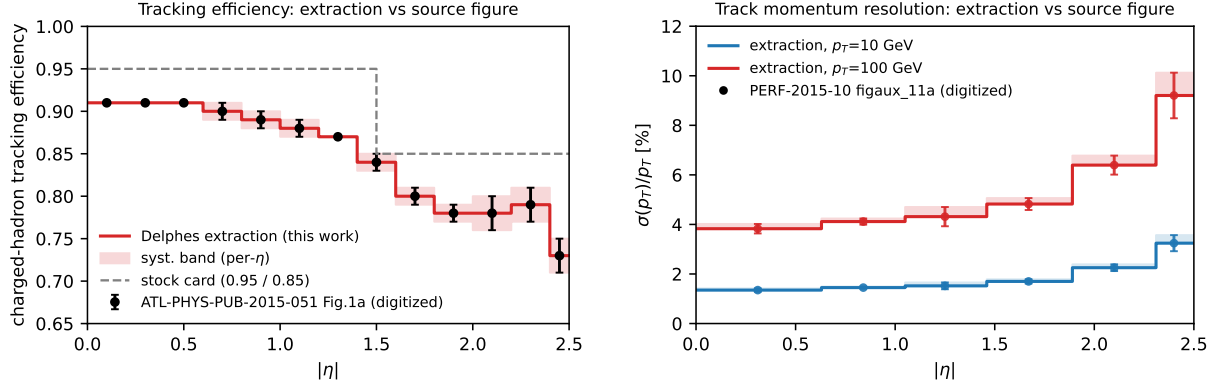


Figure 7: ATLAS charged-hadron tracking blocks (overlay on digitized points, source figures cited). Left: tracking efficiency vs. η , ATL-PHYS-PUB-2015-051 Fig. 1a. Right: track momentum resolution vs. η , PERF-2015-10 figaux_11a. Remaining blocks (electron/muon/photon identification, flavour tagging) are added to this appendix as their overlays are produced.